Creating a container and port mapping it.

- Installing and checking docker

```
C:\WINDOWS\system32\cmd.

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Bhavesh>docker version
Client:
 Version:           29.2.1
 API version:       1.53
 Go version:        go1.25.6
 Git commit:        a5c7197
 Built:             Mon Feb  2 17:20:16 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.65.0 (221669)
 Engine:
  Version:          29.2.1
  API version:      1.53 (minimum version 1.44)
  Go version:       go1.25.6
  Git commit:       6bc6209
  Built:            Mon Feb  2 17:17:24 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.2.1
  GitCommit:        dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc:
  Version:          1.3.4
  GitCommit:        v1.3.4-0-gd6d73eb8
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

- **Creating docker file**

1. Create the folder for container

2. Create the Dockerfile
   Code for the file
   FROM nginx:latest
   COPY index.html /usr/share/nginx/html/
   EXPOSE 80
   CMD ["nginx", "-g", "daemon off;"]

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Bhavesh>cd C:\docker-test

C:\docker-test>notepad index.html
```

3. Then create simple html code to run inside the container :

```
<!DOCTYPE html>
<html>
<head><title>Hello from Bhavesh</title></head>
<body><h1>Hello from Bhavesh</h1></body>
</html>
```

4. Then run command to build container from image

```
Command Prompt

C:\docker-test>docker build -t my-nginx-app .
[+] Building 8.4s (8/8) FINISHED                                                                    doc
ker:desktop-linux
 => [internal] load build definition from Dockerfile
            0.0s
 => => transferring dockerfile: 144B
            0.0s
 => [internal] load metadata for docker.io/library/nginx:latest
            3.7s
 => [auth] library/nginx:pull token for registry-1.docker.io
            0.0s
 => [internal] load .dockerignore
            0.0s
 => => transferring context: 2B
            0.0s
 => [internal] load build context
            0.0s
 => => transferring context: 169B
            0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
            4.5s
 => => resolve docker.io/library/nginx:latest@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
            0.0s
 => => sha256:5e815e07e5699b40479214a6a2a30d647495d99cd0f253ee82f528f4814469ef 1.40kB / 1.40kB
            0.6s
 => => sha256:587e3d84dbb5b5fc406b2b292318c9a446e72c144ad849b5ef8755f5037e8704 402B / 402B
            0.8s
 => => sha256:3189680c601f46244f1706d0d197ddb415d9bb754236c042acffc76eeda37d39 1.21kB / 1.21kB
            1.0s
 => => sha256:cde7a05ae42831ee510e8948b80b25c297a1080875a3479c55a65f1e620fdb73 955B / 955B
```

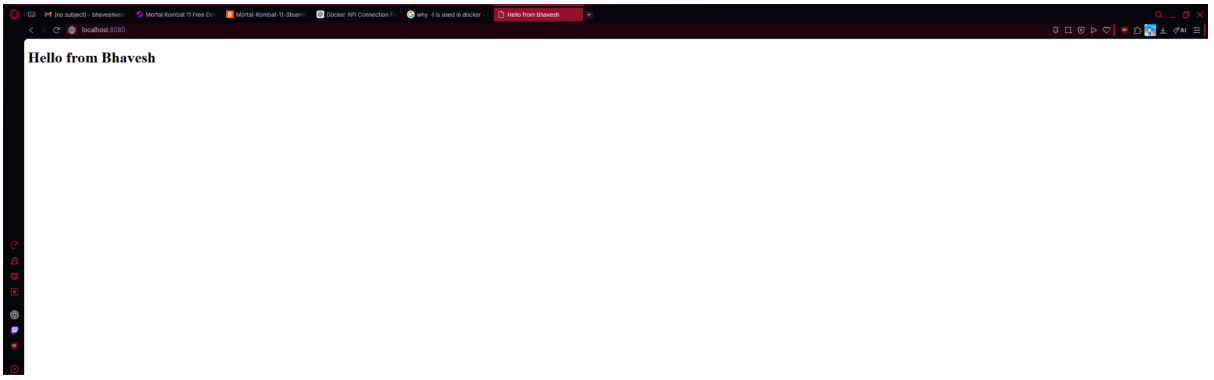
5. Then run the container and map the port .

```
Command Prompt

C:\docker-test>docker run -d -p 8080:80 my-nginx-app
7612bd46b4a1128c74a12b583f7f51aa8a2b8552e7ca7e5f44ad8339d6a99d78

C:\docker-test>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
        NAMES
7612bd46b4a1   my-nginx-app   "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes   0.0.0.0:8080->80/tcp, [::]:8080->8
0/tcp   suspicious_archimedes
```

6. Then check output on browser by running http://localhost:8080



Container Should be running

7. Now stop and delete the instance

```
Command Prompt

C:\docker-test>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS
        NAMES
7612bd46b4a1   my-nginx-app   "/docker-entrypoint.…"   3 minutes ago   Up 3 minutes   0.0.0.0:8080->80/tcp, [::]:8080->8
0/tcp   suspicious_archimedes

C:\docker-test>docker stop 7612
7612

C:\docker-test>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\docker-test>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS                     PORTS     NAMES
7612bd46b4a1   my-nginx-app   "/docker-entrypoint.…"   4 minutes ago   Exited (0) 21 seconds ago            suspicious_
archimedes

C:\docker-test>docker rm 7612
7612

C:\docker-test>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\docker-test>
```